# Myanmar Earthquake & Dam Risk Report

## 1. Executive Summary

This report covers 9,973 earthquake events in Myanmar from 1970-03-10 to 2026-10-18. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.040 with magnitude of completeness Mc=4.7.

**Total Events:** 9,973

**Date Range:** 1970-03-10 to 2026-10-18

**Max Magnitude:** M7.7

**Mean Depth:** 25.8 km

**b-value:** 1.040

**a-value:** 7.705

**Mc (completeness):** 4.7

## 2. Earthquake Map

## 3. Seismic Clustering Analysis

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## 4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

**Critical Risk:** 44 dams

**High Risk:** 116 dams

**Moderate Risk:** 45 dams

**Low Risk:** 49 dams

**Top 30 Highest Risk Dams:**

| Name | River | PGA(g) | Dist Fault | Risk | Grade |
|---|---|---|---|---|---|
| Dam near 21.74N 95.85E | nan | 0.2718 | 7.4 km | 10.0 | Critical |
| Dam near 22.09N 95.86E | nan | 0.2995 | 5.7 km | 10.0 | Critical |
| Dam near 21.77N 95.89E | nan | 0.3257 | 2.7 km | 10.0 | Critical |
| Dam near 21.47N 95.87E | nan | 0.2815 | 6.0 km | 9.7 | Critical |
| Dam near 20.85N 96.03E | nan | 0.4813 | 8.3 km | 8.8 | Critical |
| Meikhtila | nan | 0.2412 | 8.2 km | 8.8 | Critical |
| Dam near 21.34N 95.81E | nan | 0.2263 | 12.2 km | 8.8 | Critical |
| Natkar | nan | 0.2443 | 7.9 km | 8.7 | Critical |
| Dam near 21.37N 95.80E | nan | 0.2207 | 13.4 km | 8.7 | Critical |
| Dam near 20.72N 95.90E | nan | 0.2665 | 5.6 km | 8.7 | Critical |
| Dam near 21.04N 95.83E | nan | 0.2270 | 10.9 km | 8.7 | Critical |
| Dam near 21.05N 95.83E | nan | 0.2265 | 11.0 km | 8.7 | Critical |
| Dam near 20.47N 96.08E | nan | 0.5000 | 9.9 km | 8.6 | Critical |
| Dam near 20.60N 95.90E | nan | 0.2464 | 7.4 km | 8.6 | Critical |
| Taungnyo | Taungnyo | 0.3054 | 1.2 km | 8.6 | Critical |
| Nat Thar Taw | Nat Thar Taw | 0.1903 | 20.2 km | 8.5 | Critical |
| Myotha | Ku Htaw | 0.1849 | 20.6 km | 8.5 | Critical |
| Thaphan Chaung | Thaphan Chaung | 0.2197 | 11.5 km | 8.2 | Critical |
| Zeedaw | Nyaung Oat | 0.2075 | 15.3 km | 8.2 | Critical |
| Dam near 17.55N 96.37E | nan | 0.1022 | 9.6 km | 8.2 | Critical |
| Male-Nattaung | Male-Nattaung | 0.2360 | 18.6 km | 8.1 | Critical |
| Dam near 20.73N 95.83E | nan | 0.2110 | 13.6 km | 7.9 | Critical |
| Nyaunggone | nan | 0.2104 | 13.7 km | 7.9 | Critical |
| Paunggadaw | Paunggadaw | 0.1488 | 33.9 km | 7.9 | Critical |
| Kun Chaung | Kun Chaung | 0.4644 | 0.9 km | 7.7 | Critical |
| Kintat | Mu | 0.1669 | 21.0 km | 7.7 | Critical |
| North Pinle | Pinle | 0.1427 | 36.1 km | 7.7 | Critical |
| Phyu Chaung | Pyu Chaung | 0.4707 | 1.3 km | 7.6 | Critical |
| South Pinle | Pinle | 0.1383 | 38.1 km | 7.6 | Critical |
| Kabaung | Kabaung Chaung | 0.3335 | 9.2 km | 7.6 | Critical |

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## 5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

**K (productivity):** 34.92

**c (time offset):** 16.408 hours

**p (decay exponent):** 0.772

**Expected M>=3 (7d):** 3

**Expected M>=3 (30d):** 16

**Expected M>=3 (90d):** 46

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

## 6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit